

Python Math Module: Comprehensive Notes

Essential Mathematics, Functions, and Mathematical Representation in Python

1. Understanding Modules in Python

A **module** in Python is a file containing Python code—such as definitions, functions, and classes—designed to be reused across different programs. Modules allow developers to logically organize related code into separate files, promoting modularity, reusability, and maintainability.

Famous 5–6 Python Built-in Modules

- math:** Provides access to mathematical functions defined by the C standard.
- random:** Generates pseudo-random numbers for various distributions and selections.
- datetime:** Supplies classes for manipulating dates and times.
- os:** Offers operating system-dependent functionality (file paths, directories).
- sys:** Provides access to system-specific parameters, functions, and I/O handles.
- json:** Facilitates parsing and serialization of JSON data.

2. The import Statement

To use any module in Python, you bring it into your current scope using the `import` keyword. Common import patterns:

- Standard Import:** `import math` → Access via dot notation (e.g., `math.sqrt(16)`).
- Import with Alias:** `import math as m` → Access via alias (e.g., `m.sqrt(16)`).
- Selective Import:** `from math import sqrt, pi` → Import specific attributes directly.
- Import All:** `from math import *` → Imports all public names.

3. The math Module Functions (Category-Wise)

A. Number-Theoretic and Representation Functions

Python Function	Description	Mathematical Representation
<code>math.ceil(x)</code>	Smallest integer $\geq x$	$\lceil x \rceil$
<code>math.floor(x)</code>	Largest integer $\leq x$	$\lfloor x \rfloor$
<code>math.trunc(x)</code>	Truncates decimal towards zero	$\text{trunc}(x)$
<code>math.fabs(x)</code>	Absolute value (float)	$ x $
<code>math.factorial(x)</code>	Factorial of integer x	$x! = x \times (x - 1) \times \cdots \times 1$
<code>math.gcd(a, b)</code>	Greatest common divisor	$\text{gcd}(a, b)$

B. Power, Exponential, and Logarithmic Functions

Python Function	Description	Mathematical Representation
<code>math.sqrt(x)</code>	Square root of x	$\sqrt{x}$
<code>math.pow(x, y)</code>	x raised to power y	x^y
<code>math.exp(x)</code>	Exponential e^x	e^x
<code>math.log(x, [base])</code>	Logarithm to given base	$\log_{\text{base}}(x)$
<code>math.log2(x)</code>	Base-2 logarithm	$\log_2(x)$
<code>math.log10(x)</code>	Base-10 logarithm	$\log_{10}(x)$

C. Trigonometric and Inverse Trigonometric Functions

Python Function	Description	Mathematical Representation
<code>math.sin(x)</code>	Sine of x radians	$\sin(x)$
<code>math.cos(x)</code>	Cosine of x radians	$\cos(x)$
<code>math.tan(x)</code>	Tangent of x radians	$\tan(x)$
<code>math.asin(x)</code>	Arc sine of x	$\arcsin(x)$
<code>math.acos(x)</code>	Arc cosine of x	$\arccos(x)$
<code>math.atan(x)</code>	Arc tangent of x	$\arctan(x)$

D. Hyperbolic and Inverse Hyperbolic Functions

Python Function	Description	Mathematical Representation
<code>math.sinh(x)</code>	Hyperbolic sine	$\sinh(x) = \frac{e^x - e^{-x}}{2}$
<code>math.cosh(x)</code>	Hyperbolic cosine	$\cosh(x) = \frac{e^x + e^{-x}}{2}$
<code>math.tanh(x)</code>	Hyperbolic tangent	$\tanh(x) = \frac{\sinh(x)}{\cosh(x)}$
<code>math.asinh(x)</code>	Inverse hyperbolic sine	$\operatorname{asinh}(x) = \ln(x + \sqrt{x^2 + 1})$
<code>math.acosh(x)</code>	Inverse hyperbolic cosine	$\operatorname{acosh}(x) = \ln(x + \sqrt{x^2 - 1})$
<code>math.atanh(x)</code>	Inverse hyperbolic tangent	$\operatorname{atanh}(x) = \frac{1}{2} \ln\left(\frac{1+x}{1-x}\right)$

E. Special Constants and Angle Conversion

Python Function/Constant	Description	Mathematical Representation
<code>math.pi</code>	Mathematical constant π	$\pi \approx 3.14159...$
<code>math.e</code>	Mathematical constant e	$e \approx 2.71828...$
<code>math.degrees(x)</code>	Radians to degrees	$\text{degrees} = x \times \frac{180}{\pi}$
<code>math.radians(x)</code>	Degrees to radians	$\text{radians} = x \times \frac{\pi}{180}$